

Design Thinking



"Design thinking is a human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success."

—Tim Brown



What is Design Thinking?

Design thinking, or human-centered design, is a way of uncovering creative solutions to messy challenges. It starts with deeply understanding your user, then rapidly prototyping and iterating to create a desirable solution.

Have you heard of design thinking
before this course?

Design Thinking at Work



- The *Embrace* organization started as a class project in the Entrepreneurial Design for Extreme Affordability class in 2007.
- The original design brief asked the student team to build an infant incubator, to be used in the hospitals and clinics of Nepal, that is **less expensive** and **easier to maintain** than traditional Western incubators.

By reframing the problem...

From

'Create a cheaper and easier-to-maintain
incubator'

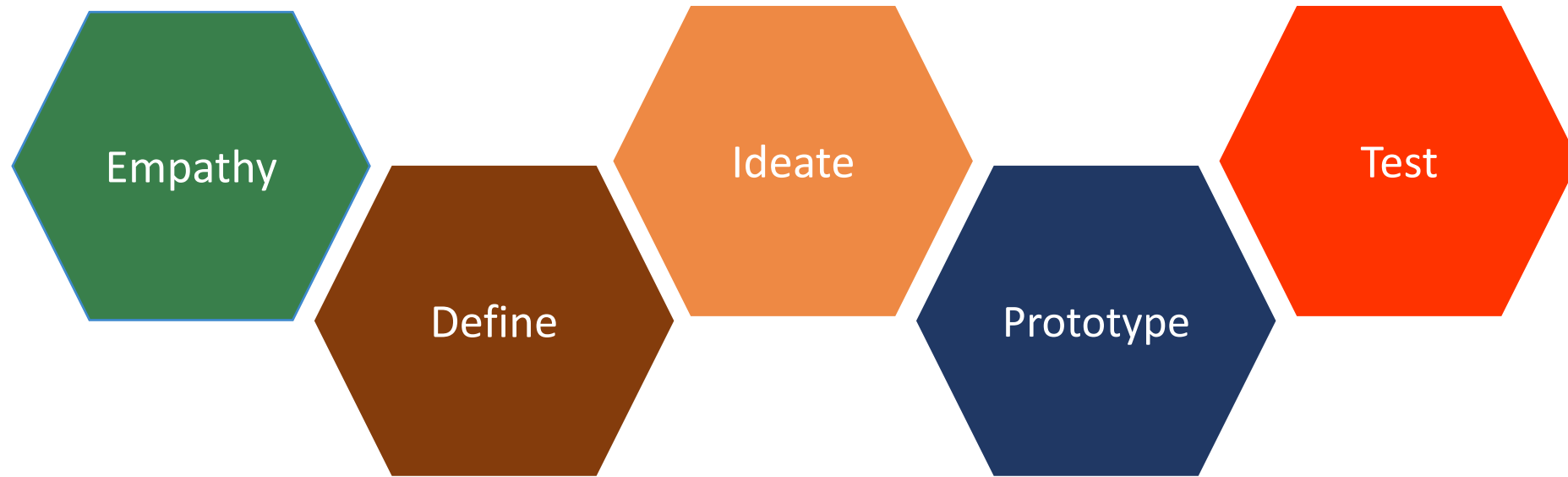
To

'Keeping babies warm'

Students created an **innovative
infant warmer**
that costs less than 1% of a
traditional incubator.



5 Modes of Design Thinking



Empathy

Empathy is the centerpiece of a human-centered design process. The Empathize mode is the work you do to understand people, within the context of your design challenge. It is your effort to understand the way they do things and why, their physical and emotional needs, how they think about world, and what is meaningful to them.

- **Observe** - View users and their behavior in the context of their lives.
- **Engage** - Interact and interview users through both scheduled and less formal encounters.
- **Immerse** - Experience what your user experiences.

Empathy



TOOLS

- ☐ Internal interviews
- ☐ Client interviews
- ☐ Empathy building activities
- ☐ Desktop research

How did

drybar®

use

Empathy

- Before launching Drybar, co-founder Alli Webb recognized that many women wanted great hair without the high cost and time commitment of a full-service salon.
- She personally experienced this frustration and validated it by providing in-home blowout services, gaining direct insights from clients.
- Their tagline, “**No cuts. No color. Just blowouts.**”, reflected their focus on simplifying the experience.
- By prioritizing empathy, Drybar revolutionized the salon industry, turning a simple service into a premium, yet accessible, self-care ritual



was solving the wrong problem

Despite inventing the digital camera, Kodak defined its business around film rather than the broader consumer need: **capturing and sharing memories**. They **resisted** digital photography because they saw it as a **threat** to film sales rather than an **evolution of consumer behavior**. By the time they adapted, competitors had taken over the market.

Define

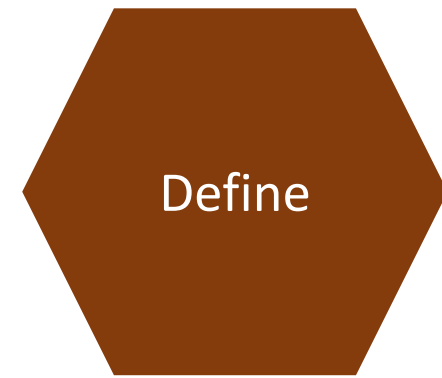
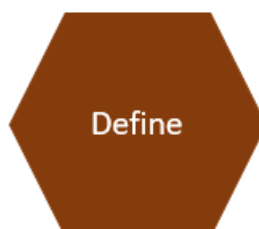
“Framing the right problem is the only way to create the right solution.”

USER : Develop an understanding of the type of person you are designing for

NEEDS : Synthesize information (gathered in empathy mode) to discover connections and patterns to identify the needs of the user.

INSIGHTS : Express insights you developed through the synthesis of information you have gathered through empathy and research work

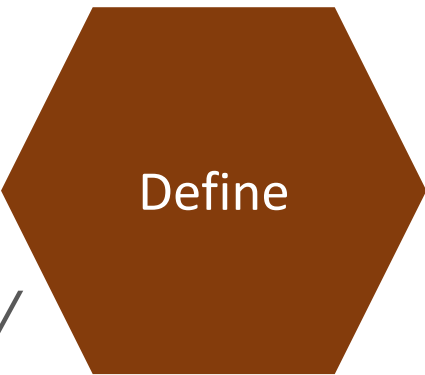
Articulate : A point-of-view (POV) by combining these three elements – user, need, and insight – as a meaningful and **actionable problem statement** that will drive the rest of your design work.



TOOLS

- ☐ Journey mapping
- ☐ Personas
- ☐ Needs sorting
- ☐ Problem statement

Point of View Statement



Define

- **We Met** *name, important descriptors*
- **We were Surprised to Learn** *an observation that was surprising / telling / insightful*
- **This Might Mean** *an inference; what might the above insight mean for his/her life?*
- **It would be Game Changing if** *what state of being would change his/her life?*

POV Example

- **We Met** *Alycia, mother of a new baby in Nepal.*
- **We were Surprised to Learn** *that she did not have access to the hospitals where the incubators were located.*
- **This Might Mean** *it is too expensive for her to travel from her rural neighborhood, so her premature baby does not have access to healthcare.*
- **It would be Game Changing if** *Alycia could more easily access quality healthcare for her baby.*

How did



Define

Empathize/Observation:

- It began with personal frustration. Dyson noticed that his traditional bagged vacuum cleaner lost suction as the bag filled.
- He observed that traditional vacuum cleaners lost suction power as the bags became clogged.

Define:

- He defined the core problem: loss of suction in vacuum cleaners due to bag clogging.
- He then specified that a solution would have to maintain constant suction, and not rely on bags.



Ideate

“It’s not about coming up with the ‘right’ idea, it’s about generating the broadest range of possibilities.”

- Ideate is where you transition from identifying problems to idea generation by “going wide” in terms of concepts and outcomes.
- It is a mode of ‘**flaring**’ rather than ‘focus’.
- Step beyond obvious solutions and thus increase the innovation potential of your solution set
- Combining your conscious and unconscious mind, and rational thoughts with imagination.
- Defer judgment – separate the generation of ideas from the evaluation of ideas : **From “No, but ..” to “Yes, and ..”**

Ideate



TOOLS

- ☐ “How might we” prompts
- ☐ Brainstorming
- ☐ Using analogies as inspiration
- ☐ Visualize the vote

Empathy

Define

Ideate

How did



Ideate

Challenge: Early on, Airbnb struggled with low customer engagement and trust issues in booking strangers' homes.

Ideation Approach:

- Airbnb's founders conducted deep brainstorming sessions and explored radical ideas to make home rentals more appealing.
- They engaged in crazy 8s (sketching eight ideas in eight minutes) and storyboarding to visualize potential solutions.
- A key insight from ideation was that high-quality photos of listings significantly improved trust and bookings. This led Airbnb to offer professional photography for hosts, which became a game-changer.

Prototype

- To communicate - If a picture is worth a thousand words, a prototype is worth a thousand pictures.
- To start a conversation - A prototype is an opportunity to have another, directed conversation with a user.
- To fail quickly and cheaply - Committing as few resources as possible to each idea means less time and money invested up front.
- To test possibilities - Staying low-res allows you to pursue many different ideas without committing to a direction too early on.

Prototype



TOOLS

- ☐ Sketching
- ☐ Storyboarding
- ☐ Wireframing
- ☐ Interactive prototype

Empathy

Define

Ideate

Prototype

How to Prototype

Prototype



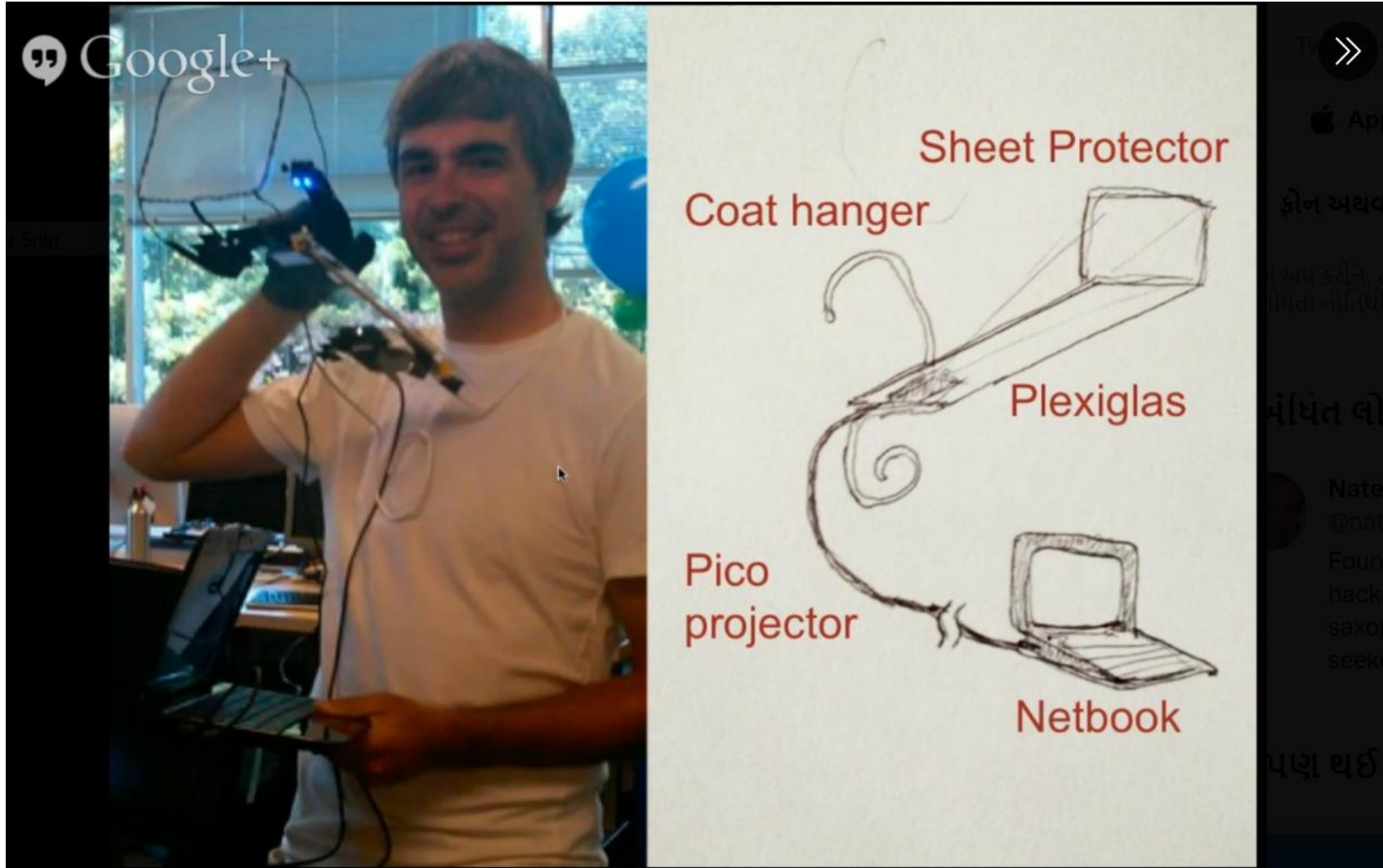
TOOLS

- ☐ Sketching
- ☐ Storyboarding
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- ☐ Interactive prototype

- Start building - Even if you aren't sure what you're doing, the act of picking up some materials (post-its, tape, and found objects are a good way to start!) will be enough to get you going.
- Don't spend too long on one prototype - Let go before you find yourself getting too emotionally attached to any one prototype.
- ID a variable - Identify what's being tested with each prototype. A prototype should answer a particular question when tested. That said, don't be blind to the other tangential understanding you can gain as someone responds to a prototype.
- Build with the user in mind - What do you hope to test with the user? What sorts of behavior do you expect? Answering these questions will help focus

Prototype Example

Prototype



Nate Aune
@natea

The first Google Glass prototype was built w/ coat hanger, sheet protector, plexiglas, pico projector and netbook.

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2 વાઈક્સ



How Many Times Should You Try?

by Anna Vital

3

times Steven Spielberg
was rejected by University
of Southern California, after which
he dropped out to become a director



26

times Tim Ferris's book
The 4 Hour Work Week was
rejected by publishers



162



times
the author of this infographic
made searches for this data

104



times
an average couple has
sex before conceiving

300



times
founder of Pandora.com
approached investors
before he got funding

400



times
a *National Geographic*
photographer shoots before
getting the final image



1,009



times
Colonel Sanders who just
started KFC was turned down
when he tried selling his
fried chicken recipe

5,126



times
James Dyson created failed
prototypes of his vacuum
cleaner before succeeding

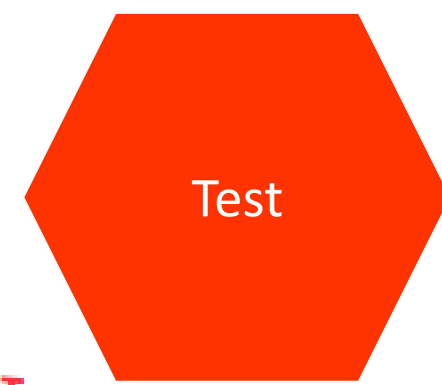
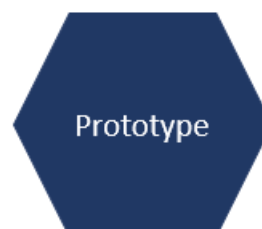
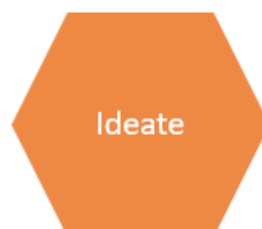
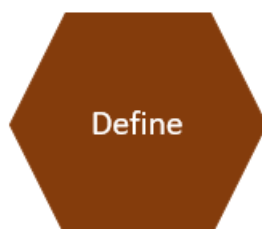
So how many
times will
you try?



Prototype

Test

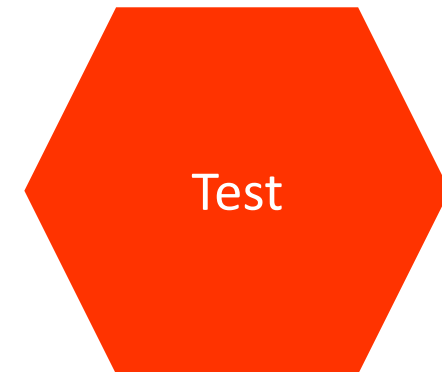
- The Test mode is when you solicit feedback, about the prototypes you have created, from your users
- Testing is the chance to refine our solutions and make them better.
- It is another iterative mode.
- Prototype as if you know you are right, but **test as if you know you are wrong.**



TOOLS

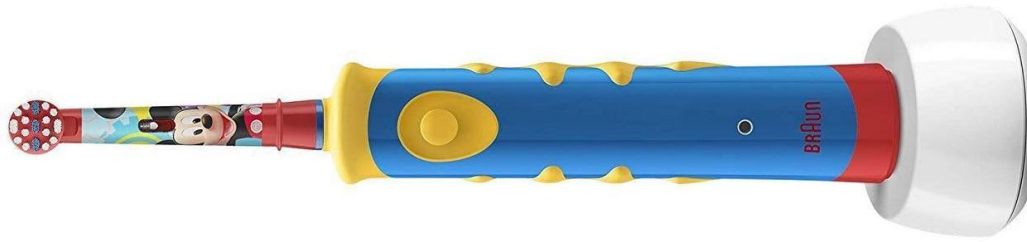
- ☐ Co-creation
- ☐ Concept testing
- ☐ Usability testing
- ☐ Pilot testing

Why Test



- To refine prototypes and solutions.
 - Testing informs the next iterations of prototypes.
 - Sometimes this means going back to the drawing board.
- To learn more about your user.
 - Testing is another opportunity to build empathy through observation and engagement—it often yields unexpected insights.
- To refine your POV.
 - Sometimes testing reveals that not only did you not get the solution right, but also that you failed to frame the problem correctly.

How did



Test

Team went into people's actual homes, stood in their bathrooms, and watched kids brush their teeth with the new prototypes.

Testing Observes Behavior, Not Just Function

- During testing, they observed that kids do not have the fine motor skills of adults. When kids tried to use the skinny, shrunken-down adult toothbrushes, they lacked control.
- By handing the kids the prototypes with the fat, squishy handles, the designers immediately saw the kids gain control. The fat handle perfectly accommodated their "power fist" grip.
- By testing the physical object in the actual environment (the bathroom sink), the unspoken needs became visible.

In Design Thinking, you **BUILD** to **THINK**, and you **TEST** to **LEARN**

Wrap-up

- The uncreative mind can spot wrong answers, but it takes a very creative mind to spot **wrong questions**.
- Ensure you have the right **problem statement** before you start solving for it.
- Instead of trying to think outside the box, **find more descriptions of the box**.
- Practicing design thinking requires **intentional** practice. You aren't an expert...yet.
- The next time you are in a brainstorm session practice divergent thinking. Instead of saying "no, but..." try saying "**yes, and...**"
- Question your **assumptions**.